
Prof. Ryan Cotterell, Prof. Mrinmaya Sachan, Prof. Florian Tramèr

Assignment 1

Goerkem Kadir Solun, 25-936-154

Collaborators

1. Collaborator 1 Name, Collaborator 1 Legi

Declaration

I, Goerkem Kadir Solun, confirm that I have written this assignment independently, without any direct copying or paraphrasing from other students' work. I have acknowledged everyone I consulted during my research. I understand that while consultation and comparison of solutions are permitted, the final written submission must be my own work.

I am aware that any form of plagiarism is a violation of academic integrity. If it is determined that I have engaged in academic dishonesty, I accept and understand that the university's standard measures for plagiarism will be applied.

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Question 1: Representation Surgery

a) Linear Guarding vs. Arbitrary Classifiers

No. By Definition 2, the statement that $\mathbf{X}$ *linearly guards* Z only guarantees that the trivially attainable loss L_τ is optimal over the set of all **affine transformations** (linear predictors). Mathematically, it ensures that $\mathbb{E}[\mathcal{L}(f(\mathbf{X}), Z)] \geq L_\tau$ holds for any affine predictor $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$. However, the classifier $c : \mathbf{x} \mapsto c(\mathbf{x}) \in \mathcal{Z}$ proposed in the question is an *arbitrary* classifier, meaning it is not restricted to being linear. Because linear guarding only constrains linear probes, a more complex, non-linear classifier such as a multi-layer perceptron could potentially draw non-linear decision boundaries to extract the concept information Z still embedded within $\mathbf{X}$. Consequently, such a non-linear classifier could successfully achieve a loss lower than the trivially attainable loss L_τ .

b) Class-Conditional Mean Equality Implies Linear Guarding

If each class-conditional mean satisfies $\mathbb{E}[\mathbf{X} \mid Z = i] = \mathbb{E}[\mathbf{X}]$, then $\mathbf{X}$ linearly guards Z w.r.t. the cross-entropy loss $\mathcal{L}$. To prove that $\mathbf{X}$ linearly guards Z , we must show that for any affine predictor $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ (where $\mathbf{A} \in \mathbb{R}^{k \times d}$ and $\mathbf{b} \in \mathbb{R}^k$), the expected loss is bounded by the trivially attainable loss L_τ :

$$\mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] \geq L_\tau \quad (1)$$

We can rewrite the expected loss by conditioning on the class Z :

$$\mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] = \mathbb{E}_Z \left[\mathbb{E}_{\mathbf{X} \mid Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z) \mid Z] \right] \quad (2)$$

The cross-entropy loss $\mathcal{L}(\mathbf{y}, Z = i) = -\ln \frac{\exp(y_i)}{\sum_{j=1}^k \exp(y_j)}$ is convex w.r.t. its input logits $\mathbf{y}$. Because $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$ is an affine, the composition $\mathcal{L}(\mathbf{A}\mathbf{x} + \mathbf{b}, Z = i)$ remains convex w.r.t. $\mathbf{x}$. Applying Jensen's inequality gives:

$$\mathbb{E}_{\mathbf{X} \mid Z=i} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, i)] \geq \mathcal{L}(\mathbb{E}_{\mathbf{X} \mid Z=i} [\mathbf{A}\mathbf{X} + \mathbf{b}], i) = \mathcal{L}(\mathbf{A} \mathbb{E}[\mathbf{X} \mid Z = i] + \mathbf{b}, i) \quad (3)$$

Use the fact that class-conditional means equal the global mean, $\mathbb{E}[\mathbf{X} \mid Z = i] = \mathbb{E}[\mathbf{X}]$.

$$\mathbb{E}_{\mathbf{X} \mid Z=i} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, i)] \geq \mathcal{L}(\mathbf{A} \mathbb{E}[\mathbf{X}] + \mathbf{b}, i) \quad (4)$$

Substituting this lower bound back into the outer expectation over Z gives:

$$\mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] \geq \mathbb{E}_Z [\mathcal{L}(\mathbf{A} \mathbb{E}[\mathbf{X}] + \mathbf{b}, Z)] \quad (5)$$

Notice that $\mathbf{b}' \triangleq \mathbf{A} \mathbb{E}[\mathbf{X}] + \mathbf{b}$ is simply a constant vector in $\mathbb{R}^k$, independent of $\mathbf{X}$. Thus, the right side of the inequality evaluates the loss of a *constant* predictor $f(\mathbf{x}) = \mathbf{b}'$.

The trivially attainable loss L_τ is the infimum of the expected loss over all constant predictors:

$$L_\tau = \inf_{\mathbf{c} \in \mathbb{R}^k} \mathbb{E}_Z [\mathcal{L}(\mathbf{c}, Z)] \quad (6)$$

Since $\mathbf{b}'$ is a constant in $\mathbb{R}^k$, its expected loss must be greater than or equal to the infimum:

$$\mathbb{E}_Z [\mathcal{L}(\mathbf{b}', Z)] \geq L_\tau \quad (7)$$

Combining the inequalities, we have shown that for any affine predictor $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$:

$$\mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(f(\mathbf{X}), Z)] \geq L_\tau \quad (8)$$

Therefore, by Definition 2, $\mathbf{X}$ linearly guards Z w.r.t. the cross-entropy loss $\mathcal{L}$.

c) **Boundedness and Symmetry of the Cross-Entropy Gradient**

Let $\mathbf{x}$ be an observed representation, and $\mathbf{y} \triangleq \mathbf{A}\mathbf{x} + \mathbf{b}$.

All partial derivatives of the cross-entropy loss w.r.t. $\mathbf{y}$ are bounded. Recall the definition of the cross-entropy loss for logit vector $\mathbf{y}$ and true concept class $Z = c$:

$$\mathcal{L}(\mathbf{y}, Z = c) = -\ln \left(\frac{\exp(y_c)}{\sum_{j=1}^k \exp(y_j)} \right) = -y_c + \ln \left(\sum_{j=1}^k \exp(y_j) \right) \quad (9)$$

We compute the partial derivative of the loss w.r.t. an arbitrary logit y_m . **Case 1** ($m = c$):

$$\frac{\partial}{\partial y_c} \mathcal{L}(\mathbf{y}, Z = c) = -1 + \frac{\frac{\partial}{\partial y_c} \sum_{j=1}^k \exp(y_j)}{\sum_{j=1}^k \exp(y_j)} \quad (10)$$

$$= -1 + \frac{\exp(y_c)}{\sum_{j=1}^k \exp(y_j)} = -1 + \text{softmax}(\mathbf{y})_c \quad (11)$$

Since the softmax output lies in $(0, 1)$ for any $\mathbf{y}$, we have $-1 < \frac{\partial}{\partial y_c} \mathcal{L}(\mathbf{y}, Z = c) < 0$.

Case 2 ($m \neq c$):

$$\frac{\partial}{\partial y_m} \mathcal{L}(\mathbf{y}, Z = c) = 0 + \frac{\frac{\partial}{\partial y_m} \sum_{j=1}^k \exp(y_j)}{\sum_{j=1}^k \exp(y_j)} \quad (12)$$

$$= \frac{\exp(y_m)}{\sum_{j=1}^k \exp(y_j)} = \text{softmax}(\mathbf{y})_m \quad (13)$$

We have $0 < \frac{\partial}{\partial y_m} \mathcal{L}(\mathbf{y}, Z = c) < 1$. Because partial derivatives fall within the interval $(-1, 1)$, they are uniformly bounded.

For any $j_1, j_2 \neq i$, we have $\frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_1) = \frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_2) \neq 0$. We find the partial derivative of the loss w.r.t. y_i when the true class is $Z = j$ with $j \neq i$. Using the derivation from above, the partial derivatives are the i -th component of the softmax output:

$$\frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_1) = \frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_2) = \frac{\exp(y_i)}{\sum_{l=1}^k \exp(y_l)} \quad (14)$$

Both expressions are equal. Furthermore, because the exponential function e^x is positive for any real x , we have $\exp(y_i) > 0$. The denominator is a sum of positive exponentials, so it is greater than zero. Therefore:

$$\frac{\exp(y_i)}{\sum_{l=1}^k \exp(y_l)} > 0 \quad (15)$$

Thus, we have shown that:

$$\frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_1) = \frac{\partial}{\partial y_i} \mathcal{L}(\mathbf{y}, Z = j_2) \neq 0 \quad (16)$$

d) Linear Guarding Implies Equal Class-Conditional Means

Because $\mathbf{X}$ linearly guards Z , the expected loss for any affine predictor $f(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{b}$ is bounded by the trivially attainable loss L_τ achieved by the constant predictor $\mathbf{b}^*$. Thus,

$$J(\mathbf{A}, \mathbf{b}) \triangleq \mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] \quad (17)$$

attains its minimum at $\mathbf{A} = \mathbf{0}$ and $\mathbf{b} = \mathbf{b}^*$. By the first-order optimality, the gradient of J w.r.t. $\mathbf{A}$, evaluated at $\mathbf{A} = \mathbf{0}, \mathbf{b} = \mathbf{b}^*$:

$$\nabla_{\mathbf{A}} \mathbb{E}_{\mathbf{X}, Z} [\mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] \Big|_{\mathbf{A}=\mathbf{0}, \mathbf{b}=\mathbf{b}^*} = \mathbf{0} \quad (18)$$

Move the derivative inside the expectation to evaluate the gradient. Let $\mathbf{y} = \mathbf{A}\mathbf{X} + \mathbf{b}$.

$$\frac{\partial \mathcal{L}}{\partial A_{ij}} = \frac{\partial \mathcal{L}}{\partial y_i} \frac{\partial y_i}{\partial A_{ij}} = \frac{\partial \mathcal{L}}{\partial y_i} X_j. \quad (19)$$

In part c we proved that the partial derivatives of the cross-entropy loss w.r.t. $\mathbf{y}$ are bounded between -1 and 1 . Because the gradient w.r.t $\mathbf{y}$ is uniformly bounded, the Dominated Convergence Theorem allows us to interchange expectation and differentiation:

$$\nabla_{\mathbf{A}} J(\mathbf{A}, \mathbf{b}) = \mathbb{E}_{\mathbf{X}, Z} [\nabla_{\mathbf{A}} \mathcal{L}(\mathbf{A}\mathbf{X} + \mathbf{b}, Z)] = \mathbb{E}_{\mathbf{X}, Z} [\nabla_{\mathbf{y}} \mathcal{L}(\mathbf{y}, Z) \mathbf{X}^\top] \quad (20)$$

From part c, for any logit y_m the partial derivative of the loss is $\frac{\partial \mathcal{L}}{\partial y_m} = \text{softmax}(\mathbf{y})_m - \mathbb{1}(Z = m)$. Let $\mathbf{p}(\mathbf{y})$ be the vector of softmax probabilities and let $\tilde{\mathbf{Z}}$ be the one-hot encoding of Z .

$$\nabla_{\mathbf{y}} \mathcal{L}(\mathbf{y}, Z) = \mathbf{p}(\mathbf{y}) - \tilde{\mathbf{Z}}. \quad (21)$$

Evaluating the gradient w.r.t. $\mathbf{A}$ at the optimum $\mathbf{A} = \mathbf{0}, \mathbf{b} = \mathbf{b}^*$ so $\mathbf{y} = \mathbf{b}^*$ yields:

$$\mathbb{E}_{\mathbf{X}, Z} [(\mathbf{p}(\mathbf{b}^*) - \tilde{\mathbf{Z}}) \mathbf{X}^\top] = \mathbf{0} \quad (22)$$

Restricting to a single class i , this implies

$$\mathbb{E}_{\mathbf{X}, Z} [(p_i(\mathbf{b}^*) - \mathbb{1}(Z = i)) \mathbf{X}] = \mathbf{0}. \quad (23)$$

By linearity of expectation,

$$p_i(\mathbf{b}^*) \mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbf{X} \mathbb{1}(Z = i)] \quad (24)$$

To determine $p_i(\mathbf{b}^*)$, recall that $\mathbf{b}^*$ minimizes the trivially attainable loss $L_\tau = \mathbb{E}_Z [\mathcal{L}(\mathbf{b}, Z)]$. First-order optimality w.r.t. $\mathbf{b}$ gives:

$$\nabla_{\mathbf{b}} \mathbb{E}_Z [\mathcal{L}(\mathbf{b}, Z)] = \mathbb{E}_Z [\nabla_{\mathbf{b}} \mathcal{L}(\mathbf{b}, Z)] = \mathbb{E}_Z [\mathbf{p}(\mathbf{b}) - \tilde{\mathbf{Z}}] = \mathbf{0} \quad (25)$$

Hence, at the optimum $\mathbf{b}^*$, the softmax probabilities must match the marginal class probabilities:

$$\mathbf{p}(\mathbf{b}^*) = \mathbb{E}_Z [\tilde{\mathbf{Z}}], \quad \text{and thus} \quad p_i(\mathbf{b}^*) = \mathbb{P}(Z = i). \quad (26)$$

Substituting $p_i(\mathbf{b}^*) = \mathbb{P}(Z = i)$ back into (24):

$$\mathbb{P}(Z = i) \mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbf{X} \mathbb{1}(Z = i)] \quad (27)$$

By the definition of conditional expectation, $\mathbb{E}[\mathbf{X} \mathbb{1}(Z = i)] = \mathbb{E}[\mathbf{X} | Z = i] \mathbb{P}(Z = i)$, so

$$\mathbb{P}(Z = i) \mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbf{X} | Z = i] \mathbb{P}(Z = i). \quad (28)$$

Since $\mathbb{P}(Z = i) \neq 0$ by assumption, we can divide both sides by $\mathbb{P}(Z = i)$ to obtain

$$\mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbf{X} | Z = i]. \quad (29)$$

This holds for every class i .

e) **Zero Cross-Covariance Characterizes Equal Class-Conditional Means**

The cross-covariance between $\mathbf{X}$ and $\tilde{\mathbf{Z}}$ is

$$\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbb{E}[\mathbf{X} \tilde{\mathbf{Z}}^\top] - \mathbb{E}[\mathbf{X}] \mathbb{E}[\tilde{\mathbf{Z}}]^\top. \quad (30)$$

Its j^{th} column is the covariance of the vector $\mathbf{X}$ with the scalar $\tilde{Z}_j$:

$$\text{Cov}(\mathbf{X}, \tilde{Z}_j) = \mathbb{E}[\mathbf{X} \tilde{Z}_j] - \mathbb{E}[\mathbf{X}] \mathbb{E}[\tilde{Z}_j]. \quad (31)$$

Since $\tilde{Z}_j = \mathbb{1}(Z = j)$, we have $\mathbb{E}[\tilde{Z}_j] = \mathbb{P}(Z = j)$ and, by the definition of conditional expectation, $\mathbb{E}[\mathbf{X} \tilde{Z}_j] = \mathbb{E}[\mathbf{X} | Z = j] \mathbb{P}(Z = j)$. Substituting gives:

$$\text{Cov}(\mathbf{X}, \tilde{Z}_j) = \left(\mathbb{E}[\mathbf{X} | Z = j] - \mathbb{E}[\mathbf{X}] \right) \mathbb{P}(Z = j) \quad (32)$$

Forward direction Assume $\mathbb{E}[\mathbf{X} | Z = j] = \mathbb{E}[\mathbf{X}]$ for every j . Then the parenthesised term vanishes, so $\text{Cov}(\mathbf{X}, \tilde{Z}_j) = \mathbf{0}$ for every column j , and therefore $\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}$.

Backward direction Assume $\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}$. Every column must then be zero, so for each j :

$$\left(\mathbb{E}[\mathbf{X} | Z = j] - \mathbb{E}[\mathbf{X}] \right) \mathbb{P}(Z = j) = \mathbf{0}. \quad (33)$$

Since $\mathbb{P}(Z = j) \neq 0$ by assumption, the directions give $\mathbb{E}[\mathbf{X} | Z = j] = \mathbb{E}[\mathbf{X}] \iff \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}$.

f) **Linear Guarding under an Affine Map**

A representation linearly guards Z if and only if its cross-covariance with the one-hot encoded concept vector $\tilde{\mathbf{Z}}$ is zero. Applied to the transformed representation $r(\mathbf{X}) = \mathbf{P}\mathbf{X} + \mathbf{b}$:

$$r(\mathbf{X}) \text{ linearly guards } Z \iff \Sigma_{r(\mathbf{X}), \tilde{\mathbf{Z}}} = \mathbf{0}. \quad (34)$$

Adding the constant vector $\mathbf{b}$ only shifts the mean and does not change covariance, so

$$\text{Cov}(\mathbf{P}\mathbf{X} + \mathbf{b}, \tilde{\mathbf{Z}}) = \text{Cov}(\mathbf{P}\mathbf{X}, \tilde{\mathbf{Z}}). \quad (35)$$

For any constant matrix $\mathbf{P}$ and random vectors $\mathbf{X}, \mathbf{Y}$, $\text{Cov}(\mathbf{P}\mathbf{X}, \mathbf{Y}) = \mathbf{P} \text{Cov}(\mathbf{X}, \mathbf{Y})$.

$$\text{Cov}(\mathbf{P}\mathbf{X}, \tilde{\mathbf{Z}}) = \mathbf{P} \text{Cov}(\mathbf{X}, \tilde{\mathbf{Z}}) = \mathbf{P} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}. \quad (36)$$

Hence $\Sigma_{r(\mathbf{X}), \tilde{\mathbf{Z}}} = \mathbf{P} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}$. Combining with the guarding characterisation above yields:

$$r(\mathbf{X}) \text{ linearly guards } Z \iff \mathbf{P} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}. \quad (37)$$

g) **Decoupling the Distortion under a Centered Representation**

Define the distortion induced by the affine map $r(\mathbf{X}) = \mathbf{P}\mathbf{X} + \mathbf{b}$ as $D(\mathbf{P}, \mathbf{b}) \triangleq \mathbb{E}[\|\mathbf{P}\mathbf{X} + \mathbf{b} - \mathbf{X}\|^2]$.

Part 1: If $\mathbb{E}[\mathbf{X}] = \mathbf{0}$, then $D(\mathbf{P}, \mathbf{b}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2] + \|\mathbf{b}\|^2$. Group the terms with $\mathbf{X}$:

$$D(\mathbf{P}, \mathbf{b}) = \mathbb{E}[\|(\mathbf{P}\mathbf{X} - \mathbf{X}) + \mathbf{b}\|^2]. \quad (38)$$

Use $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2\langle \mathbf{u}, \mathbf{v} \rangle$ with $\mathbf{u} = \mathbf{P}\mathbf{X} - \mathbf{X}$ and $\mathbf{v} = \mathbf{b}$, and expectation linearity:

$$D(\mathbf{P}, \mathbf{b}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2] + \|\mathbf{b}\|^2 + 2\mathbb{E}[\langle \mathbf{P}\mathbf{X} - \mathbf{X}, \mathbf{b} \rangle]. \quad (39)$$

Since $\mathbf{b}$ is deterministic, $\mathbb{E}[\|\mathbf{b}\|^2] = \|\mathbf{b}\|^2$. Linearity of the inner product and of expectation gives:

$$\mathbb{E}[\langle \mathbf{P}\mathbf{X} - \mathbf{X}, \mathbf{b} \rangle] = \langle \mathbb{E}[\mathbf{P}\mathbf{X} - \mathbf{X}], \mathbf{b} \rangle = \langle (\mathbf{P} - \mathbf{I}) \mathbb{E}[\mathbf{X}], \mathbf{b} \rangle. \quad (40)$$

Since $\mathbb{E}[\mathbf{X}] = \mathbf{0}$, this reduces to $\langle \mathbf{0}, \mathbf{b} \rangle = 0$, and the cross term vanishes, leaving:

$$D(\mathbf{P}, \mathbf{b}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2] + \|\mathbf{b}\|^2. \quad (41)$$

Part 2: Minimizing $D(\mathbf{P}, \mathbf{b})$ subject to a constraint on $\mathbf{P}$ is equivalent to minimizing $D(\mathbf{P}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2]$. From part 1, when $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ the distortion decouples into two non-negative terms:

$$D(\mathbf{P}, \mathbf{b}) = \underbrace{\mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2]}_{\text{depends only on } \mathbf{P}} + \underbrace{\|\mathbf{b}\|^2}_{\text{depends only on } \mathbf{b}}. \quad (42)$$

The constraint is only on $\mathbf{P}$, so $\mathbf{b}$ can be chosen freely. Since $\|\mathbf{b}\|^2 \geq 0$, the unconstrained optimum is $\mathbf{b}^* = \mathbf{0}$. Substituting back, the minimization reduces to

$$D(\mathbf{P}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2]. \quad (43)$$

h) Optimal Guarding Projection under Whitened Representations

Assume $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ and $\Sigma_{\mathbf{X}, \mathbf{X}} = \mathbf{I}$. Show the following satisfies Eq. (7) and minimizes Eq. (8).

$$\mathbf{P}^* = \mathbf{I} - \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}^+ \quad (44)$$

Part 1: $\mathbf{P}^*$ satisfies the guarding constraint. Substitute $\mathbf{P}^*$ into the constraint:

$$\mathbf{P}^* \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} - \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}^+ \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}. \quad (45)$$

By the property of the pseudoinverse, $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$ for any matrix $\mathbf{A}$, so

$$\mathbf{P}^* \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} - \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}. \quad (46)$$

Part 2: $\mathbf{P}^*$ minimizes the distortion. From part (g), we minimize $D(\mathbf{P}) = \mathbb{E}[\|\mathbf{P}\mathbf{X} - \mathbf{X}\|^2] = \mathbb{E}[\|(\mathbf{P} - \mathbf{I})\mathbf{X}\|^2]$ subject to $\mathbf{P} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}$. Applying Hint 1 together with $\Sigma_{\mathbf{X}, \mathbf{X}} = \mathbf{I}$:

$$D(\mathbf{P}) = \text{tr}((\mathbf{P} - \mathbf{I}) \Sigma_{\mathbf{X}, \mathbf{X}} (\mathbf{P} - \mathbf{I})^\top) = \text{tr}((\mathbf{P} - \mathbf{I})(\mathbf{P} - \mathbf{I})^\top) = \|\mathbf{P} - \mathbf{I}\|_F^2. \quad (47)$$

The problem reduces to minimizing $\|\mathbf{P} - \mathbf{I}\|_F^2$ subject to constraint $\mathbf{P} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}$. Writing $\mathbf{P}$ by its rows $\mathbf{p}_i^\top$, the constraint forces each row to satisfy $\mathbf{p}_i^\top \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} = \mathbf{0}^\top$, each $\mathbf{p}_i$ lies in the orthogonal complement of $V \triangleq \text{colsp}(\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}})$. The Frobenius objective decouples row-wise:

$$\|\mathbf{P} - \mathbf{I}\|_F^2 = \sum_i \|\mathbf{p}_i - \mathbf{e}_i\|^2, \quad (48)$$

so the optimum is found by independently projecting each $\mathbf{e}_i$ onto $V^\perp$: $\mathbf{p}_i^* = \Pi_{V^\perp} \mathbf{e}_i$. Stacking the rows using $\Pi_{V^\perp}^\top = \Pi_{V^\perp}$ yields $\mathbf{P}^* = \Pi_{V^\perp}$. By Hint 2 applied to $\mathbf{A} = \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}$:

$$\mathbf{P}^* = \Pi_{V^\perp} = \mathbf{I} - \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}^+. \quad (49)$$

i) Extension to Arbitrary Mean and Covariance via Whitening

Drop the simplifying assumptions and allow $\mathbb{E}[\mathbf{X}] \neq \mathbf{0}$ and $\Sigma_{\mathbf{X}, \mathbf{X}} \neq \mathbf{I}$. Define the whitening matrix $\mathbf{W} \triangleq (\Sigma_{\mathbf{X}, \mathbf{X}}^{1/2})^+$.

Whitening the representation. Let $\mathbf{Y} \triangleq \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}])$. Then

$$\mathbb{E}[\mathbf{Y}] = \mathbf{W}(\mathbb{E}[\mathbf{X}] - \mathbb{E}[\mathbf{X}]) = \mathbf{0}, \quad \Sigma_{\mathbf{Y}, \mathbf{Y}} = \mathbf{W} \Sigma_{\mathbf{X}, \mathbf{X}} \mathbf{W}^\top = \mathbf{I} \quad (50)$$

on the support of $\mathbf{X} - \mathbb{E}[\mathbf{X}]$. By translation invariance and linearity of covariance in its first argument:

$$\Sigma_{\mathbf{Y}, \tilde{\mathbf{Z}}} = \text{Cov}(\mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]), \tilde{\mathbf{Z}}) = \mathbf{W} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}. \quad (51)$$

Applying part (h) in the whitened space. Since $\mathbf{Y}$ now satisfies the assumptions of part (h), the optimal guarding projection in whitened coordinates is

$$\mathbf{P}_{\mathbf{Y}}^* = \mathbf{I} - \Sigma_{\mathbf{Y}, \tilde{\mathbf{Z}}} \Sigma_{\mathbf{Y}, \tilde{\mathbf{Z}}}^+ = \mathbf{I} - \Pi_{\mathbf{W}\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}}, \quad (52)$$

so the guarded whitened representation is $\hat{\mathbf{Y}} = \mathbf{P}_{\mathbf{Y}}^* \mathbf{Y}$.

Mapping back to the original space. Undo the whitening and centering by applying $\mathbf{W}^+ = \Sigma_{\mathbf{X}, \mathbf{X}}^{1/2}$ and adding back $\mathbb{E}[\mathbf{X}]$:

$$r(\mathbf{X}) = \mathbf{W}^+ \hat{\mathbf{Y}} + \mathbb{E}[\mathbf{X}] \quad (53)$$

$$= \mathbf{W}^+ \left(\mathbf{I} - \Pi_{\mathbf{W}\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}} \right) \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]) + \mathbb{E}[\mathbf{X}] \quad (54)$$

$$= \mathbf{W}^+ \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]) - \mathbf{W}^+ \Pi_{\mathbf{W}\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}} \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]) + \mathbb{E}[\mathbf{X}]. \quad (55)$$

The product $\mathbf{W}^+ \mathbf{W} = \Sigma_{\mathbf{X}, \mathbf{X}}^{1/2} (\Sigma_{\mathbf{X}, \mathbf{X}}^{1/2})^+$ is the orthogonal projection onto $\text{colsp}(\Sigma_{\mathbf{X}, \mathbf{X}}^{1/2})$, which contains $\mathbf{X} - \mathbb{E}[\mathbf{X}]$ almost surely, so $\mathbf{W}^+ \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]) = \mathbf{X} - \mathbb{E}[\mathbf{X}]$. The means cancel, giving the final form:

$$r(\mathbf{X}) = \mathbf{X} - \mathbf{W}^+ \Pi_{\mathbf{W}\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}} \mathbf{W}(\mathbf{X} - \mathbb{E}[\mathbf{X}]), \quad (56)$$

where $\Pi_{\mathbf{W}\Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}}} = (\mathbf{W} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}})(\mathbf{W} \Sigma_{\mathbf{X}, \tilde{\mathbf{Z}}})^+$ is the orthogonal projection in the whitened space onto the subspace encoding concept information.

Question 2: Properties of Finite-state Language Models

a) The Expected String Length

To compute the expected string length μ , we must translate the probability distribution over strings into matrix operations. $\mu \triangleq \sum_{\mathbf{y} \in \Sigma^*} p(\mathbf{y})|\mathbf{y}|$.

In a probabilistic finite-state automaton (PFSA) $\mathcal{A} = (\Sigma, Q, \delta, \lambda, \rho)$, we can group all strings by their length k . The probability of the model generating *any* string of length k is given by the sum of the weights of all paths of length k . Using the transition matrix $\mathbf{T}$, the initial state probabilities λ , and the stopping probabilities ρ , the total probability for all strings of length k is given by $\lambda^\top \mathbf{T}^k \rho$.

$$\mu = \sum_{k=0}^{\infty} k (\lambda^\top \mathbf{T}^k \rho) = \lambda^\top \left(\sum_{k=0}^{\infty} k \mathbf{T}^k \right) \rho \quad (57)$$

To compute this in $\mathcal{O}(|Q|^3)$, we can call the `allsum` algorithm as a black box. The `allsum` algorithm evaluates the sum of weights for all possible paths, computing $\sum_{k=0}^{\infty} \mathbf{W}^k = (\mathbf{I} - \mathbf{W})^{-1}$ for a given weight matrix $\mathbf{W}$.

To make our equation match the `allsum` formulation, we can construct a new augmented automaton using a block matrix structure.

We define a modified PFSA over an expanded state space Q' where each original state is duplicated, meaning $|Q'| = 2|Q|$. This allows the transition matrix to accumulate the path length k as it is raised to higher powers.

We define a new transition block matrix $\mathbf{M} \in \mathbb{R}^{2|Q| \times 2|Q|}$, a new initial state vector $\lambda' \in \mathbb{R}^{2|Q|}$, and a new stopping vector $\rho' \in \mathbb{R}^{2|Q|}$ as follows:

$$\mathbf{M} = \begin{pmatrix} \mathbf{T} & \mathbf{T} \\ \mathbf{0} & \mathbf{T} \end{pmatrix}, \quad \lambda' = \begin{pmatrix} \lambda \\ \mathbf{0} \end{pmatrix}, \quad \rho' = \begin{pmatrix} \mathbf{0} \\ \rho \end{pmatrix} \quad (58)$$

When we raise the block matrix $\mathbf{M}$ to the power of k , the diagonal blocks remain $\mathbf{T}^k$, while the upper-right block accumulates the sum of the products. For any integer $k \geq 0$, we have:

$$\mathbf{M}^k = \begin{pmatrix} \mathbf{T}^k & k \mathbf{T}^k \\ \mathbf{0} & \mathbf{T}^k \end{pmatrix} \quad (59)$$

If we apply the `allsum` algorithm to this new augmented automaton, it computes the sum of the weights over all paths of all lengths:

$$\text{allsum}(\mathbf{M}, \lambda', \rho') = \sum_{k=0}^{\infty} \lambda'^\top \mathbf{M}^k \rho' \quad (60)$$

Substituting block definitions into this equation shows the length coefficient:

$$\lambda'^\top \mathbf{M}^k \rho' = (\lambda^\top \quad \mathbf{0}^\top) \begin{pmatrix} \mathbf{T}^k & k \mathbf{T}^k \\ \mathbf{0} & \mathbf{T}^k \end{pmatrix} \begin{pmatrix} \mathbf{0} \\ \rho \end{pmatrix} \quad (61)$$

$$= (\lambda^\top \mathbf{T}^k \quad \lambda^\top (k \mathbf{T}^k)) \begin{pmatrix} \mathbf{0} \\ \rho \end{pmatrix} \quad (62)$$

$$= \lambda^\top k \mathbf{T}^k \rho \quad (63)$$

Summing this extraction over all $k \in [0, \infty)$ yields the expected length μ :

$$\sum_{k=0}^{\infty} \lambda'^{\top} \mathbf{M}^k \rho' = \sum_{k=0}^{\infty} \lambda^{\top} k \mathbf{T}^k \rho = \mu \quad (64)$$

The Algorithm

1. Construct the block matrix $\mathbf{M}$ alongside vectors λ' and ρ' .
2. Call the standard `allsum` function, passing $(\mathbf{M}, \lambda', \rho')$ as the arguments.
3. Return the scalar output as μ .

Proof of Time Complexity

- Constructing the augmented matrix $\mathbf{M}$ and the vectors involves allocating memory and copying the elements of $\mathbf{T}$, which takes $\mathcal{O}(|Q|^2)$ time.
- The `allsum` algorithm evaluates the infinite sum by calculating the inverse $(\mathbf{I} - \mathbf{M})^{-1}$, or by utilizing Gaussian elimination to solve the equivalent linear system. Matrix inversion and solving linear systems both inherently have a cubic time complexity relative to the size of the matrix.
- Since our augmented matrix $\mathbf{M}$ has dimensions $2|Q| \times 2|Q|$, the `allsum` algorithm will execute in $\mathcal{O}((2|Q|)^3)$ time.
- Because constants can be dropped in Big-O notation, $\mathcal{O}(8|Q|^3)$ simplifies directly to $\mathcal{O}(|Q|^3)$.

The $\mathcal{O}(|Q|^2)$ setup time is strictly bounded and dominated by the $\mathcal{O}(|Q|^3)$ `allsum` operation. Thus, the total overall runtime of the algorithm is proven to be $\mathcal{O}(|Q|^3)$.

Alternative Implementation To elaborate further on the inherent properties of finite-state language models, we can also bypass the block matrix construction to yield a slightly more memory-efficient algorithm that still meets the $\mathcal{O}(|Q|^3)$ requirement. We know from geometric series properties of matrices that $\sum_{k=0}^{\infty} \mathbf{T}^k = (\mathbf{I} - \mathbf{T})^{-1}$. If we evaluate the infinite sum $\sum_{k=0}^{\infty} k \mathbf{T}^k$ by factoring out $\mathbf{T}$ and rewriting it as a matrix derivative, we arrive at:

$$\sum_{k=0}^{\infty} k \mathbf{T}^k = \mathbf{T}(\mathbf{I} - \mathbf{T})^{-2} = (\mathbf{I} - \mathbf{T})^{-1} \mathbf{T} (\mathbf{I} - \mathbf{T})^{-1} \quad (65)$$

This means the mean string length can also be written purely algebraically as $\mu = \lambda^{\top} (\mathbf{I} - \mathbf{T})^{-1} \mathbf{T} (\mathbf{I} - \mathbf{T})^{-1} \rho$. We can compute this directly using a sequence of linear algebraic operations:

1. **Solve** the linear system $(\mathbf{I} - \mathbf{T})\mathbf{v}_1 = \rho$ to find $\mathbf{v}_1$. Time complexity: $\mathcal{O}(|Q|^3)$.
2. **Multiply** the transition matrix to find $\mathbf{v}_2 = \mathbf{T}\mathbf{v}_1$. Time complexity: $\mathcal{O}(|Q|^2)$.
3. **Solve** the second linear system $(\mathbf{I} - \mathbf{T})\mathbf{v}_3 = \mathbf{v}_2$ to find $\mathbf{v}_3$. Time complexity: $\mathcal{O}(|Q|^3)$.
4. **Compute** the final dot product $\mu = \lambda^{\top} \mathbf{v}_3$. Time complexity: $\mathcal{O}(|Q|)$.

The sum of these sequential execution times is $\mathcal{O}(|Q|^3) + \mathcal{O}(|Q|^2) + \mathcal{O}(|Q|^3) + \mathcal{O}(|Q|)$, which perfectly asymptotes to the required $\mathcal{O}(|Q|^3)$ runtime. Both methods are formally correct, though the block matrix approach directly fulfills the hint to deploy the existing `allsum` operation as an undisturbed black box.

b) The Variance of String Lengths

The variance of the string length is defined as $\sigma^2 = \sum_{\mathbf{y} \in \Sigma^*} p(\mathbf{y})(|\mathbf{y}| - \mu)^2$. Using the linearity of expectation:

$$\sigma^2 = \mathbb{E}[|\mathbf{y}|^2] - \mu^2 \quad (66)$$

Sum over all path lengths k to compute the second moment $\mathbb{E}[|\mathbf{y}|^2]$:

$$\mathbb{E}[|\mathbf{y}|^2] = \sum_{k=0}^{\infty} k^2 (\lambda^\top \mathbf{T}^k \rho) = \lambda^\top \left(\sum_{k=0}^{\infty} k^2 \mathbf{T}^k \right) \rho \quad (67)$$

Decompose the k^2 into $k(k-1) + k$ to separate the sum:

$$\sum_{k=0}^{\infty} k^2 \mathbf{T}^k = \sum_{k=0}^{\infty} k(k-1) \mathbf{T}^k + \sum_{k=0}^{\infty} k \mathbf{T}^k \quad (68)$$

These sums map to the derivatives of the geometric matrix series $\sum_{k=0}^{\infty} \mathbf{T}^k = (\mathbf{I} - \mathbf{T})^{-1}$:

1. The first gives the sum for k : $\sum_{k=0}^{\infty} k \mathbf{T}^k = \mathbf{T}(\mathbf{I} - \mathbf{T})^{-2}$.
2. The second gives the sum for $k(k-1)$: $\sum_{k=0}^{\infty} k(k-1) \mathbf{T}^k = 2\mathbf{T}^2(\mathbf{I} - \mathbf{T})^{-3}$.

$$\sum_{k=0}^{\infty} k^2 \mathbf{T}^k = 2\mathbf{T}^2(\mathbf{I} - \mathbf{T})^{-3} + \mathbf{T}(\mathbf{I} - \mathbf{T})^{-2} \quad (69)$$

$$= (\mathbf{I} - \mathbf{T})^{-3} (2\mathbf{T}^2 + \mathbf{T}(\mathbf{I} - \mathbf{T})) = (\mathbf{I} - \mathbf{T})^{-3} (\mathbf{T}^2 + \mathbf{T}) \quad (70)$$

Thus, the second moment is $\mathbb{E}[|\mathbf{y}|^2] = \lambda^\top (\mathbf{T}^2 + \mathbf{T})(\mathbf{I} - \mathbf{T})^{-3} \rho$. Finally, the variance is $\sigma^2 = \lambda^\top (\mathbf{T}^2 + \mathbf{T})(\mathbf{I} - \mathbf{T})^{-3} \rho - \mu^2$.

Algorithm

1. **Compute μ :** Use the algorithm in (a) to find the mean string length.
2. **Initialize:** Define the base vector $\mathbf{v}_0 = \rho$.
3. **Solve First System:** $(\mathbf{I} - \mathbf{T})\mathbf{v}_1 = \mathbf{v}_0$ for $\mathbf{v}_1$. (This computes $(\mathbf{I} - \mathbf{T})^{-1}\rho$).
4. **Solve Second System:** $(\mathbf{I} - \mathbf{T})\mathbf{v}_2 = \mathbf{v}_1$ for $\mathbf{v}_2$. (This computes $(\mathbf{I} - \mathbf{T})^{-2}\rho$).
5. **Solve Third System:** $(\mathbf{I} - \mathbf{T})\mathbf{v}_3 = \mathbf{v}_2$ for $\mathbf{v}_3$. (This computes $(\mathbf{I} - \mathbf{T})^{-3}\rho$).
6. **Compute Polynomial Matrix:** Calculate the matrix $\mathbf{U} = \mathbf{T}^2 + \mathbf{T}$.
7. **Compute Vector:** Calculate the product $\mathbf{w} = \mathbf{U} \mathbf{v}_3$.
8. **Compute Second Moment:** Calculate the dot product $\mathbb{E}[|\mathbf{y}|^2] = \lambda^\top \mathbf{w}$.
9. **Compute Variance:** Return the variance as $\sigma^2 = \mathbb{E}[|\mathbf{y}|^2] - \mu^2$.

Proof of Time Complexity

- **Step 1:** Computing the expected length μ was proven in (a) to take $\mathcal{O}(|Q|^3)$.
- **Steps 3, 4, 5:** Solving a linear system of size $|Q| \times |Q|$ has a worst-case time complexity of $\mathcal{O}(|Q|^3)$. We solve three linear systems sequentially making the time taken is $3 \times \mathcal{O}(|Q|^3)$, which asymptotically is $\mathcal{O}(|Q|^3)$.
- **Step 6:** Multiplying two square matrices of size $|Q| \times |Q|$ to calculate $\mathbf{T}^2$ requires $\mathcal{O}(|Q|^3)$ time. Adding $\mathbf{T}$ requires $\mathcal{O}(|Q|^2)$ additions.

- **Step 7:** Multiplying the $|Q| \times |Q|$ matrix $\mathbf{U}$ by the $|Q| \times 1$ vector $\mathbf{v}_3$ takes $\mathcal{O}(|Q|^2)$.
- **Step 8:** The dot product of two vectors of size $|Q|$ executes in $\mathcal{O}(|Q|)$.
- **Step 9:** A single scalar subtraction operation happens in $\mathcal{O}(1)$ time.

Total sum of the time complexity:

$$\mathcal{O}(|Q|^3) + \mathcal{O}(|Q|^3) + \mathcal{O}(|Q|^3) + \mathcal{O}(|Q|^2) + \mathcal{O}(|Q|) + \mathcal{O}(1) \quad (71)$$

The highest order term dominates the growth rate, so the algorithm runs in time $\mathcal{O}(|Q|^3)$.

c) Entropy of a Deterministic Finite-State Language Model

Because the automaton $\mathcal{A}$ is deterministic, any string $\mathbf{y}$ corresponds to one unique path of transitions through the automaton from a given starting state.

- Let Π_q be a random variable representing the paths in $\mathcal{A}$ starting in state q .
- Let $\mathbf{H}(\Pi_q)$ denote the entropy of the distribution of paths originating from state q .

Decompose a path Π_q into its first step and the remainder of the path.

- Let τ_q be a random variable representing the outgoing transition from q or its termination.
- Because $\mathcal{A}$ is deterministic, the destination state q' is uniquely determined by τ_q . Consequently, the path Π_q is in bijection with the pair $(\tau_q, \Pi_{q'})$, so their entropies coincide: $\mathbf{H}(\Pi_q) = \mathbf{H}(\tau_q, \Pi_{q'})$.
- By the chain rule of entropy, $\mathbf{H}(x, y) = \mathbf{H}(x) + \mathbf{H}(y | x)$. Applying this to the joint variable $(\tau_q, \Pi_{q'})$ yields:

$$\mathbf{H}(\Pi_q) = \mathbf{H}(\tau_q, \Pi_{q'}) = \mathbf{H}(\tau_q) + \mathbf{H}(\Pi_{q'} | \tau_q) \quad (72)$$

Evaluate the entropy of the first step, $\mathbf{H}(\tau_q)$. The probability distribution for τ_q is defined taking the transition $q \xrightarrow{y/w} q'$ with w , and taking the termination with ρ_q . The entropy of this single step is the negative sum of $p \log p$ for all possible outcomes:

$$\mathbf{H}(\tau_q) = - \sum_{q \xrightarrow{y/w} q'} w \log w - \rho_q \log \rho_q = \xi_q \quad (73)$$

Evaluate the conditional entropy of the remaining path, $\mathbf{H}(\Pi_{q'} | \tau_q)$ represents the expected entropy of the rest of the path, given the first step τ_q . If τ_q is termination with probability ρ_q , the sequence ends, and the entropy of the remaining empty path is 0. If τ_q is a transition to state q' with probability w , the remaining path originates from q' , so its entropy is $\mathbf{H}(\Pi_{q'})$. Summing these expectations over all outgoing transitions gives:

$$\mathbf{H}(\Pi_{q'} | \tau_q) = \sum_{q \xrightarrow{y/w} q'} w \mathbf{H}(\Pi_{q'}) \quad (74)$$

We can group this sum by the destination state q' . By definition, the transition matrix entry $\mathbf{T}_{q,q'}$ is the sum of the weights w of all transitions from q to q' . Therefore:

$$\mathbf{H}(\Pi_{q'} | \tau_q) = \sum_{q' \in Q} \mathbf{T}_{q,q'} \mathbf{H}(\Pi_{q'}) \quad (75)$$

Construct the matrix equation by substituting the results:

$$\mathbf{H}(\Pi_q) = \xi_q + \sum_{q' \in Q} \mathbf{T}_{q,q'} \mathbf{H}(\Pi_{q'}) \quad (76)$$

To solve, let $\mathbf{H} \in \mathbb{R}^{|Q|}$ be a column vector where the q -th entry is $H(\Pi_q)$. We can represent the system of equations for all states as a single matrix equation:

$$\mathbf{H} = \boldsymbol{\xi} + \mathbf{T} \mathbf{H} \quad (77)$$

Solve for the path entropy vector $\mathbf{H}$ by isolating $\mathbf{H}$ using algebraic matrix operations:

$$\mathbf{H} - \mathbf{T} \mathbf{H} = \boldsymbol{\xi} = (\mathbf{I} - \mathbf{T}) \mathbf{H} \quad (78)$$

Because all eigenvalues of the transition matrix $\mathbf{T}$ have a magnitude less than 1, the matrix $(\mathbf{I} - \mathbf{T})$ is guaranteed to be invertible:

$$\mathbf{H} = (\mathbf{I} - \mathbf{T})^{-1} \boldsymbol{\xi} \quad (79)$$

The overall entropy of the language model $H(p)$ is the expected path entropy weighted by the initial state probabilities. Since $\mathcal{A}$ is deterministic with a unique initial state (so $H(\lambda) = 0$), every string $y \in \Sigma^*$ corresponds to exactly one complete path. The distribution over strings therefore coincides with the distribution over paths, giving $H(p) = H(\Pi) = \sum_{q \in Q} \lambda_q H(\Pi_q)$. Written as the dot product of the initial state distribution and the path entropy vector:

$$H(p) = \lambda^\top \mathbf{H} \quad (80)$$

Substitute the solved vector $\mathbf{H}$ into the final equation:

$$H(p) = \lambda^\top (\mathbf{I} - \mathbf{T})^{-1} \boldsymbol{\xi} \quad (81)$$

Question 3: Efficiently Simulating Bounded Stacks with RNNs

- a) The language $D(2, 3)$ consists of brackets of $k = 2$ types with maximum nesting depth $m = 3$. We encode the unclosed opening brackets using a stack whose *rightmost* symbol is the top of the stack. $[\langle_1 \langle_2]$ transitions to $[\langle_1]$ after reading $\rangle_2$.
- Reading a closing bracket $\rangle_i$: pop the top of the stack if it equals $\langle_i$; otherwise (mismatch or empty stack), reject.
 - Reading an opening bracket $\langle_i$: push it on top, unless the stack already has $m = 3$ symbols, in which case the depth bound is exceeded and we reject.

	$\gamma_1 = \langle_2$	$\gamma_2 = \langle_1 \langle_1 \langle_1$	$\gamma_3 = \langle_2 \langle_1 \langle_2$
$\rangle_2$	$\emptyset$	reject	$\langle_2 \langle_1$
$\rangle_1$	reject	$\langle_1 \langle_1$	reject
$\langle_2$	$\langle_2 \langle_2$	reject	reject

Table 1: The first row of the table represents different stack configurations, while the first column lists the newly read symbols.

- b) We construct a DFA $\mathcal{A} = (\Sigma, Q, I, F, \delta)$ that recognizes $D(k, m)$ by encoding every legal stack configuration as a state. Let $\langle = \{\langle_1, \dots, \langle_k\}$ and $\rangle = \{\rangle_1, \dots, \rangle_k\}$.

Alphabet: $\Sigma = \langle \cup \rangle$, so $|\Sigma| = 2k$.

States: $Q = \{[\gamma] \mid \gamma \in \langle^{\leq m}\} \cup \{R\}$, where $[\gamma]$ corresponds to the stack of unclosed brackets γ with $|\gamma| \leq m$, and R is an absorbing rejecting state. $A \stackrel{\text{def}}{=} [\emptyset]$ for the empty-stack state.

Initial states: $I = \{A\}$ (start with an empty stack).

Final states: $F = \{A\}$ (only the empty stack accepts).

Transition $\delta : Q \times \Sigma \rightarrow Q$: For valid prefix $\gamma \in \langle^*$ with $[\gamma] \in Q$ and $i, j \in \{1, \dots, k\}$:

1. *Valid push* (if $|\gamma| < m$): $\delta([\gamma], \langle_i) = [\gamma \langle_i]$.
2. *Valid pop*: $\delta([\gamma \langle_i], \rangle_i) = [\gamma]$.
3. *Depth overflow* (if $|\gamma| = m$): $\delta([\gamma], \langle_i) = R$.
4. *Mismatched close* (if $i \neq j$): $\delta([\gamma \langle_j], \rangle_i) = R$.
5. *Underflow*: $\delta(A, \rangle_i) = R$.
6. *Sink*: $\delta(R, s) = R$ for every $s \in \Sigma$.

These cases specify δ on every $(Q \setminus \{R\}) \times \Sigma$ pair as well as on $\{R\} \times \Sigma$, so δ is deterministic.

Correctness. A string y is in $D(k, m)$ iff (i) the brackets are balanced and (ii) the running depth $d(y_{\leq t}) \leq m$. The state $[\gamma_t]$ reached after reading $y_{\leq t}$ stores the multiset *and order* of currently unclosed brackets, so $|\gamma_t| = d(y_{\leq t})$. The transitions enforce the two conditions:

- Any depth violation triggers rule (3), sending $\mathcal{A}$ into the absorbing R .
- Any mismatched close (rule 4) or underflow (rule 5) likewise sends $\mathcal{A}$ to R .

Conversely, while $y_{\leq t} \in D(k, m)$ at every prefix, only rules (1) and (2) fire and the automaton stays inside $\{[\gamma] \mid \gamma \in \langle^{\leq m}\}$. Since $F = \{A\}$, the run accepts iff after reading the whole string the stack is empty, i.e. all opens were correctly matched. Hence $\mathcal{L}(\mathcal{A}) = D(k, m)$.

Complexity.

$$|Q| = 1 + \sum_{j=0}^m k^j = 1 + \frac{k^{m+1} - 1}{k - 1} = \mathcal{O}(k^m).$$

Because $\mathcal{A}$ is a DFA, the number of transitions is

$$|\delta| = |Q| \cdot |\Sigma| = \mathcal{O}(k^m) \cdot 2k = \mathcal{O}(k^{m+1}).$$

- c) For $D(2, 3)$ we have $k = 2$ and $m = 3$, so each hidden state lives in $\mathbb{R}^{k \cdot m} = \mathbb{R}^6$. Writing $\mathbf{e}_1 = (1, 0)^\top$ and $\mathbf{e}_2 = (0, 1)^\top$ for the canonical basis vectors of $\mathbb{R}^2$ associated with $\langle 1$ and $\langle 2$, Eq. (27) places the top of the stack in the first k -block and pads unused bottom blocks with zeros. The empty stack maps to $h([\emptyset]) = \mathbf{0}_6$, and the initial embeddings of $\gamma_1, \gamma_2, \gamma_3$ are

$$\begin{aligned} h(\gamma_1) &= h([\langle 2]) = \begin{pmatrix} \mathbf{e}_2 \\ \mathbf{0}_2 \\ \mathbf{0}_2 \end{pmatrix} = (0, 1, 0, 0, 0, 0)^\top, \\ h(\gamma_2) &= h([\langle 1 \langle 1]) = \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_1 \\ \mathbf{e}_1 \end{pmatrix} = (1, 0, 1, 0, 1, 0)^\top, \\ h(\gamma_3) &= h([\langle 2 \langle 1 \langle 2]) = \begin{pmatrix} \mathbf{e}_2 \\ \mathbf{e}_1 \\ \mathbf{e}_2 \end{pmatrix} = (0, 1, 1, 0, 0, 1)^\top. \end{aligned}$$

Re-encoding the resulting stacks from part (a) via Eq. (27) yields the embeddings:

	$\gamma_1 = \langle 2$	$\gamma_2 = \langle 1 \langle 1$	$\gamma_3 = \langle 2 \langle 1 \langle 2$
$\rangle_2$	$(0, 0, 0, 0, 0, 0)^\top$	reject	$(1, 0, 0, 1, 0, 0)^\top$
$\rangle_1$	reject	$(1, 0, 1, 0, 0, 0)^\top$	reject
$\langle 2$	$(0, 1, 0, 1, 0, 0)^\top$	reject	reject

Table 2: The first row of the table represents different stack configurations, while the first column lists the newly read symbols.

- d) Let $\mathbf{I}_k$ and $\mathbf{0}_k$ denote the $k \times k$ identity and zero matrices, and write $\mathbf{h}$ as m stacked k -blocks where block j is the j -th symbol from the top of the stack. Block 1 is the top, blocks beyond the stack depth are zero. Pushing onto the stack means *shifting all k -blocks one position down* and writing the new symbol into block 1. Popping means *shifting all k -blocks one position up* and zeroing block m . Realize this with two block-banded $km \times km$ matrices,

$$\mathbf{U}_{\text{push}} = \begin{pmatrix} \mathbf{0}_k & \mathbf{0}_k & \cdots & \mathbf{0}_k & \mathbf{0}_k \\ \mathbf{I}_k & \mathbf{0}_k & \cdots & \mathbf{0}_k & \mathbf{0}_k \\ \mathbf{0}_k & \mathbf{I}_k & \cdots & \mathbf{0}_k & \mathbf{0}_k \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_k & \mathbf{0}_k & \cdots & \mathbf{I}_k & \mathbf{0}_k \end{pmatrix}, \quad \mathbf{U}_{\text{pop}} = \begin{pmatrix} \mathbf{0}_k & \mathbf{I}_k & \mathbf{0}_k & \cdots & \mathbf{0}_k \\ \mathbf{0}_k & \mathbf{0}_k & \mathbf{I}_k & \cdots & \mathbf{0}_k \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_k & \mathbf{0}_k & \mathbf{0}_k & \cdots & \mathbf{I}_k \\ \mathbf{0}_k & \mathbf{0}_k & \mathbf{0}_k & \cdots & \mathbf{0}_k \end{pmatrix},$$

and define the recurrence-matrix function $\mathbf{U} : \bar{\Sigma} \rightarrow \mathbb{R}^{km \times km}$ by

$$\mathbf{U}(y_t) = \begin{cases} \mathbf{U}_{\text{push}} & \text{if } y_t \in \{\langle 1, \dots, \langle k\}, \\ \mathbf{U}_{\text{pop}} & \text{if } y_t \in \{\rangle_1, \dots, \rangle_k\}, \\ \mathbf{I}_{km} & \text{if } y_t = \text{EOS}. \end{cases}$$

The choice $\mathbf{U}(\text{EOS}) = \mathbf{I}_{km}$ keeps the stack representation intact at termination, since EOS is not a stack operation. The input matrix $\mathbf{V} \in \mathbb{R}^{km \times (2k+1)}$ injects the new opening bracket into block 1 only when y_t is an opening bracket, so we set

$$\mathbf{V} = \begin{pmatrix} \mathbf{I}_k \\ \mathbf{0}_k \\ \vdots \\ \mathbf{0}_k \end{pmatrix} (\mathbf{I}_k \quad \mathbf{0}_k \quad \mathbf{0}_{k \times 1}) = \begin{pmatrix} \mathbf{I}_k & \mathbf{0}_k & \mathbf{0}_{k \times 1} \\ \mathbf{0}_k & \mathbf{0}_k & \mathbf{0}_{k \times 1} \\ \vdots & \vdots & \vdots \\ \mathbf{0}_k & \mathbf{0}_k & \mathbf{0}_{k \times 1} \end{pmatrix},$$

where the columns are ordered as $\langle_1, \dots, \langle_k, \rangle_1, \dots, \rangle_k, \text{EOS}$. With this $\mathbf{V}$, $\mathbf{V}\mathbf{o}_{\langle_i} = (\mathbf{e}_i, \mathbf{0}_{(m-1)k})^\top$ and $\mathbf{V}\mathbf{o}_{\rangle_i} = \mathbf{V}\mathbf{o}_{\text{EOS}} = \mathbf{0}_{km}$.

Correspondence with $\mathcal{A}$. Suppose $\mathbf{h}_{t-1}$ encodes a non-rejecting state $q = [\langle_{i_1} \dots \langle_{i_{m'}}] \in \mathcal{Q}$ via Eq. (27). We check the three transition kinds.

Valid push. If $y_t = \langle_i$ and $m' < m$, $\mathcal{A}$ moves to $q' = [\langle_{i_1} \dots \langle_{i_{m'}} \langle_i]$. Multiplying out, $\mathbf{U}_{\text{push}} \mathbf{h}_{t-1}$ takes block j of $\mathbf{h}_{t-1}$ into block $j+1$ for $j = 1, \dots, m-1$ and zeros block 1. Adding $\mathbf{V}\mathbf{o}_{\langle_i}$ writes $\mathbf{e}_i$ into block 1. The resulting blocks are $(\mathbf{e}_i, \mathbf{e}_{i_{m'}}, \dots, \mathbf{e}_{i_1}, \mathbf{0}, \dots, \mathbf{0})$, exactly $h(q')$.

Valid pop. If $y_t = \rangle_i$ with $i = i_{m'}$, $\mathcal{A}$ moves to $q' = [\langle_{i_1} \dots \langle_{i_{m'-1}}]$. Now $\mathbf{U}_{\text{pop}} \mathbf{h}_{t-1}$ takes block j into block $j-1$ for $j = 2, \dots, m$ and zeros block m , while $\mathbf{V}\mathbf{o}_{\rangle_i} = \mathbf{0}$. The resulting blocks are $(\mathbf{e}_{i_{m'-1}}, \dots, \mathbf{e}_{i_1}, \mathbf{0}, \dots, \mathbf{0})$, exactly $h(q')$.

EOS. The DFA does not consume EOS (cf. part (b)). We simply preserve the stack via $\mathbf{U}(\text{EOS}) = \mathbf{I}_{km}$ and $\mathbf{V}\mathbf{o}_{\text{EOS}} = \mathbf{0}$, so $\mathbf{h}_t = \mathbf{h}_{t-1}$.

Invalid transitions. Pushing on a full stack ($m' = m$) loses block m (shifted out), and popping a mismatched or empty stack discards the wrong $\mathbf{e}$ or stays at $\mathbf{0}$. In either case $\mathbf{h}_t$ no longer encodes valid stack. The dynamics map alone cannot detect this. It is the recognition module of part (f) that assigns probability $< \eta$ to such continuations, so in the locally η -truncated support these strings are rejected.

Hence, on every *valid* prefix, the RNN's hidden states are in one-to-one correspondence with the states $\mathcal{A}$ visits, and the recurrence $\mathbf{h}_t = \mathbf{U}(y_t)\mathbf{h}_{t-1} + \mathbf{V}\mathbf{o}_{y_t}$ implements δ .

- e) Since $\mathbf{U}$ is now fixed, it can't depend on whether y_t should push or pop. Following the hint, double the hidden state to $\mathbb{R}^{2km}$ and perform *both* operations simultaneously in each half:

$$\mathbf{h}_t = \begin{pmatrix} \mathbf{h}_t^{(p)} \\ \mathbf{h}_t^{(q)} \end{pmatrix} \in \mathbb{R}^{2km},$$

where the top half $\mathbf{h}_t^{(p)}$ holds the result of a push and the bottom half $\mathbf{h}_t^{(q)}$ holds the result of a pop. The Heaviside activation H together with large negative entries in $\mathbf{V}$ then gates off whichever half does not match the input. We maintain the invariant that one half is non-zero and equals the correct stack encoding $h([\gamma])$ from Eq. (27). The true stack is recovered as $\mathbf{s}_t \stackrel{\text{def}}{=} \mathbf{h}_t^{(p)} + \mathbf{h}_t^{(q)}$.

Let $\mathbf{U}_{\text{push}}, \mathbf{U}_{\text{pop}} \in \mathbb{R}^{km \times km}$ be the shift matrices from part (d) and let $\mathbf{V}_{\text{ins}} \in \mathbb{R}^{km \times k}$ be the matrix that places $\mathbf{e}_i$ into block 1 when its input is $\langle_i$, i.e. $\mathbf{V}_{\text{ins}} = (\mathbf{I}_k, \mathbf{0}_k, \dots, \mathbf{0}_k)^\top$ (the first k columns of the $\mathbf{V}$ from part (d)). Define

$$\mathbf{U} = \begin{pmatrix} \mathbf{U}_{\text{push}} & \mathbf{U}_{\text{push}} \\ \mathbf{U}_{\text{pop}} & \mathbf{U}_{\text{pop}} \end{pmatrix} \in \mathbb{R}^{2km \times 2km}, \quad \mathbf{b} = -\frac{1}{2} \mathbf{1}_{2km} \in \mathbb{R}^{2km},$$

$$\mathbf{V} = \begin{pmatrix} \mathbf{V}_{\text{ins}} & -2 \mathbf{1}_{km \times k} & -2 \mathbf{1}_{km \times 1} \\ -2 \mathbf{1}_{km \times k} & \mathbf{0}_{km \times k} & -2 \mathbf{1}_{km \times 1} \end{pmatrix} \in \mathbb{R}^{2km \times (2k+1)},$$

where the $2k+1$ columns are ordered as $\langle_1, \dots, \langle_k, \rangle_1, \dots, \rangle_k, \text{EOS}$.

Why $\mathbf{U}$ is correct. Multiplying out,

$$\mathbf{U}\mathbf{h}_{t-1} = \begin{pmatrix} \mathbf{U}_{\text{push}}(\mathbf{h}_{t-1}^{(p)} + \mathbf{h}_{t-1}^{(q)}) \\ \mathbf{U}_{\text{pop}}(\mathbf{h}_{t-1}^{(p)} + \mathbf{h}_{t-1}^{(q)}) \end{pmatrix} = \begin{pmatrix} \mathbf{U}_{\text{push}}\mathbf{s}_{t-1} \\ \mathbf{U}_{\text{pop}}\mathbf{s}_{t-1} \end{pmatrix},$$

so we recover both candidate next-stacks from the true stack $\mathbf{s}_{t-1}$. One of $\mathbf{h}_{t-1}^{(p)}, \mathbf{h}_{t-1}^{(q)}$ is $\mathbf{0}$ and the other is a 0/1 encoding, hence $\mathbf{s}_{t-1} \in \{0, 1\}^{km}$ and the entries of $\mathbf{U}\mathbf{h}_{t-1}$ lie in $\{0, 1\}$.

Correspondence with $\mathcal{A}$. Suppose $\mathbf{s}_{t-1} = h(q)$ for some non-rejecting state $q = [\langle_{i_1} \cdots \langle_{i_{m'}}] \in \mathcal{Q}$.

Valid push ($y_t = \langle_i$, $m' < m$, $q' = [\langle_{i_1} \cdots \langle_{i_{m'}} \langle_i]$). The one-hot $\mathbf{o}_{y_t}$ selects column i of $\mathbf{V}$:

$$\mathbf{V}\mathbf{o}_{\langle_i} = \begin{pmatrix} \mathbf{V}_{\text{ins}} \mathbf{e}_i \\ -2 \mathbf{1}_{km} \end{pmatrix} = \begin{pmatrix} (\mathbf{e}_i, \mathbf{0}_{(m-1)k})^\top \\ -2 \mathbf{1}_{km} \end{pmatrix} \Rightarrow \mathbf{z}_t = \begin{pmatrix} \mathbf{U}_{\text{push}}\mathbf{s}_{t-1} + (\mathbf{e}_i, \mathbf{0})^\top - \frac{1}{2}\mathbf{1} \\ \mathbf{U}_{\text{pop}}\mathbf{s}_{t-1} - \frac{5}{2}\mathbf{1} \end{pmatrix}.$$

The top half: $\mathbf{U}_{\text{push}}\mathbf{s}_{t-1}$ has its block 1 zeroed out, so adding $(\mathbf{e}_i, \mathbf{0})^\top$ never causes a collision. Every entry is 0 or 1, and after subtracting $\frac{1}{2}$ they sit at $\pm\frac{1}{2}$. The Heaviside yields $h(q')$. The bottom half: each entry is $\leq 1 - \frac{5}{2} = -\frac{3}{2} < 0$, so H makes it $\mathbf{0}$. Thus $\mathbf{h}_t^{(p)} = h(q')$, $\mathbf{h}_t^{(q)} = \mathbf{0}$, and $\mathbf{s}_t = h(q')$.

Valid pop ($y_t = \rangle_i$, $i = i_{m'}$, $q' = [\langle_{i_1} \cdots \langle_{i_{m'-1}}]$). The one-hot picks column $k + i$:

$$\mathbf{V}\mathbf{o}_{\rangle_i} = \begin{pmatrix} -2 \mathbf{1}_{km} \\ \mathbf{0}_{km} \end{pmatrix} \Rightarrow \mathbf{z}_t = \begin{pmatrix} \mathbf{U}_{\text{push}}\mathbf{s}_{t-1} - \frac{5}{2}\mathbf{1} \\ \mathbf{U}_{\text{pop}}\mathbf{s}_{t-1} - \frac{1}{2}\mathbf{1} \end{pmatrix}.$$

The top half is $\leq -\frac{3}{2} < 0$, so H zeros it. The bottom half has entries in $\{-\frac{1}{2}, +\frac{1}{2}\}$ and H recovers $\mathbf{U}_{\text{pop}}\mathbf{s}_{t-1} = h(q')$. Hence $\mathbf{h}_t^{(p)} = \mathbf{0}$, $\mathbf{h}_t^{(q)} = h(q')$, and $\mathbf{s}_t = h(q')$.

EOS. The last column of $\mathbf{V}$ is $-2 \mathbf{1}_{2km}$, so both halves of $\mathbf{z}_t$ are $\leq -\frac{3}{2}\mathbf{1}$ and $\mathbf{h}_t = \mathbf{0}_{2km}$. Since the recognition module of part (f) reads $\mathbf{h}_{t-1}$ to compute the probability of $y_t = \text{EOS}$, this loss of information at $t = T$ does not affect recognition.

Invalid transitions. A push on a full stack ($m' = m$) shifts the bottom block of $\mathbf{s}_{t-1}$ off the top half (it is annihilated by $\mathbf{U}_{\text{push}}$), so $\mathbf{h}_t^{(p)}$ encodes a *different* length- m stack instead of the rejecting state of $\mathcal{A}$. Similarly, a mismatched or empty pop produces a top half that does not equal $h(q')$ for the true continuation. The dynamics of part (e) cannot detect these failures by themselves; this is the role of the output layer constructed in part (f), which assigns probability $< \eta$ to such invalid continuations and thus excludes them from the locally η -truncated support. On every valid prefix, however, the recurrence $\mathbf{h}_t = H(\mathbf{U}\mathbf{h}_{t-1} + \mathbf{V}\mathbf{o}_{y_t} + \mathbf{b})$ tracks the transitions of $\mathcal{A}$ exactly via $\mathbf{s}_t$.

- f)
- g)
- h)
- i)

Question 4: Limitations of Transformers on Simple Languages

- a)
- b)
- c)
- d)
- e)
- f)
- g)
- h)
- i)
- j)
- k)
- l)